TT-W-549D
November 24, 1975
SUPERSEDING
Fed. Spec. TT-W-549C
August 3, 1961

FEDERAL SPECIFICATION

WOOD PRESERVATIVE: AMMONIACAL COPPER ARSENATE MIXTURE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the composition and methods of analysis of an ammoniacal copper arsenate treating solution for the preservative treatment of wood according to Federal Specification TT-W-571 (6.1), and also a strong solution suitable for the brush treatment of the cut ends of treated wood products.

1.2 Classification.

1.2.1 Composition. Ammoniacal copper arsenate shall be of one composition, as specified herein.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-E-485 - Enamel, Semi-Gloss, Rust-Inhibiting.
TT-W-571 - Wood Preservation: Treating Practices.
PPP-C-96 - Cans, Metal: 28 Gage and Lighter.
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon.
PPP-P-704 - Pails, Metal: (Shipping, steel, 1 Through 12 Gallon).

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standard:

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

40 CFR 162 - Federal Insecticide, Fungicide, and Rodenticide Act.

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Wood-Preservers' Association (AWPA) Standard:

A2 - Standard Methods for Analysis of Water-Borne Preservatives and Fire-Retardant Formulations.

(Application for copies should be addressed to the American Wood-Preservers' Association, 1625 Eye Street, N.W., Washington, DC 20006.)

3. REQUIREMENTS

3.1 Properties.

3.1.1 Ammoniacal copper arsenate mixture shall consist of compounds soluble in dilute ammonia solution, and selected from the following groups each in excess of 95 percent purity on an anhydrous basis:

Divalent copper - e.g., cupric hydroxide
Trivalent arsenic - e.g., arsenic trioxide

The commercial preservative shall be labeled as to its total content of active ingredients calculated as cupric oxide (CuO) and arsenic pentoxide (As_2O_5).

3.1.2 The active ingredients of ammoniacal copper arsenate, either in a treating solution or in a concentrated solution, shall be present in the following ratio:

Copper, as CuO.....49.8 parts by weight

Arsenic, as As_2O_5.....50.2 parts by weight

The above shall be dissolved in a solution of ammonia (NH_3) in water. The weight of ammonia contained in a solution shall be from 1.5 to 2.0 times the weight of the copper oxide. To aid in dissolving the chemicals, a maximum of 1.7 percent of glacial acetic acid may be added.

3.1.3 The composition of a treating solution or of a concentrated solution shall not fall below the following limits:

	Minimum percent
Copper, as CuO.....	47.7
Arsenic, as As_2O_5.....	47.6

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3.1.4 The concentration of a treating solution shall be adjusted so as to result in a retention as specified for the product. The concentration of a strong solution shall be a minimum of 5 percent calculated as CuO and As_2O_5.

3.1.5 The net retention of preservative shall be calculated as net pounds of CuO plus As_2O_5 deposited in the wood in the proportions as limited by 3.1.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The treating solution shall be thoroughly agitated, after which a 1-quart sample shall be taken for inspection and test purposes. The contractor shall supply such samples and containers as may be required at no cost to the Government.

4.3 Tests. Tests to determine the properties of ammoniacal copper arsenate treating solutions (see 3.1) shall be made in accordance with AWPA Standard A2.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A or B. The preservative shall be furnished in 5-gallon steel containers conforming to type I, class 1 or 8 of Federal Specification PPP-P-704 or 55-gallon steel drums conforming to type I or II of Federal Specification PPP-D-729. Five-gallon containers shall be furnished with pour spouts (see 6.2).

5.1.2 Level C. The preservative shall be furnished in 5-gallon steel containers or in 55-gallon steel drums normally furnished for commercial practice (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to any special marking required, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Ammoniacal copper arsenate may be used for the preservative treatment by pressure methods of lumber, structural timbers, piles, poles, posts, and plywood in accordance with the conditions outlined in Federal Specification TT-W-571, Wood Preservation: Treating Practices. A strong solution (3.1.4) may be used for brush applications to the cut surfaces of treated products.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.1).
(c) Size of drum required for level A or level B packing (see 5.1.1).

6.3 Normally, only treated materials will be procured by the Government, and there are no Governmental plants using the bulk preservative.

6.3.1 If the material covered by this specification is procured by the Government as an end item, it should be marked in accordance with the Federal Insecticide, Fungicide, and Rondenticide Act (see 2.1).

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6.4 This specification was developed by the Forest Products Laboratory, Forest Service, U.S. Department of Agriculture, Madison, WI, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

MILITARY CUSTODIANS:

Army - ME
Air Force - 84

Coordinating activity:

Army - ME

Preparing activity:

AGR-FS

U. S. GOVERNMENT PRINTING OFFICE : 1976 - 210-814/1081

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